

Article

Crying Wolf in Cyberspace: A Cybersecurity Dynamics Study of Alarm Fatigue Attacks

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Abstract

Modern cyber–physical infrastructures rely heavily on alarm and notification systems to direct human attention when abnormal conditions occur. These mechanisms support timely and safe responses by informing operators and occupants about potential hazards. At the same time, research in human factors has shown that repeated or excessive alerts can weaken vigilance, slow reactions, and reduce confidence in warning systems. This behavioral pattern is commonly described as alarm fatigue. This paper examines how that vulnerability can be exploited intentionally. We refer to this adversarial strategy as *alarm poisoning*: the deliberate injection of false or misleading alerts in order to increase alarm pressure, erode trust in the monitoring infrastructure, and degrade organizational responsiveness over time. To study this process, we develop a stochastic Cybersecurity Dynamics model representing the interaction among attackers, defenders, alarm infrastructure, and a population of employees. Employee behavior is modeled through evolving trust and fatigue levels, while the overall system is formulated as a continuous–time Markov chain and simulated using the Gillespie Stochastic Simulation Algorithm. A Monte–Carlo campaign is used to analyze the resulting socio–technical dynamics under alternative attacker strategies. The study evaluates time–dependent trust, fatigue, and alarm–pressure trajectories, the distribution of times to behavioral collapse, and defender timing through Trust–Resilience–Agility–Mitigation (TRAM) metrics. The revised analysis also includes replication–sufficiency diagnostics, one–at–a–time sensitivity analysis, and threshold–robustness checks for the collapse criterion. The results show that false alarms with high perceived severity drive alarm pressure upward and degrade trust faster than nuisance–dominated campaigns, even when the total fake–alarm intensity is held constant across strategies. Collapse timing remains highly variable across stochastic realizations, and a non–negligible fraction of runs do not reach the collapse threshold within the simulation horizon. Sensitivity analysis indicates that the main qualitative ranking of attacker strategies is robust across most tested perturbations, with fatigue recovery and defender escalation emerging as particularly influential mechanisms. Overall, the findings support the view that alarm poisoning is a credible socio–technical attack vector and highlight the importance of rapid mitigation, robust alarm management, and human–centered defensive design in cyber–physical security systems.



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1. Introduction

Human attention is limited and can be worn down over time. Research in psychology and human factors engineering has repeatedly observed that when people are exposed to

frequent warnings, notifications, or interruptions, their level of vigilance tends to decline. Reaction times often increase, and confidence in alerting systems may weaken. In safety-critical environments such as aviation, healthcare, and industrial control, this pattern—commonly referred to as alarm fatigue—has led to ongoing efforts to redesign alarm systems through prioritization schemes, filtering mechanisms, and adaptive alerting strategies.

A similar situation appears in digital infrastructures. Contemporary buildings, data centers, and industrial facilities depend on extensive networks of sensors and automated alerting systems that signal abnormal conditions, from cybersecurity incidents to physical hazards. These systems are intended to help operators notice problems quickly and react appropriately. In practice, however, alarms can become too frequent, poorly tuned, or repeatedly inaccurate. When that happens, the warning system may gradually lose credibility and fail to attract the attention it was designed to secure.

This work explores how such limitations could be intentionally exploited. An adversary might avoid direct disruption and instead compromise monitoring components in order to generate streams of false or misleading alerts with different levels of severity. Over time, this persistent exposure may encourage occupants and operators to doubt the alerts they receive. If attention has already eroded and distrust has become widespread within the organization, a legitimate emergency might trigger a slower or weaker response than expected. We refer to this form of socio-technical manipulation as *alarm poisoning*.

Understanding these dynamics requires models that capture more than isolated technical failures. The evolution of security conditions depends on the interaction between infrastructure, defensive organizations, and the behavior of human participants. Our approach follows the Cybersecurity Dynamics perspective, where security is treated as a property that evolves over time through the interplay of attacks, defenses, and human responses.

Within this framework, we construct a stochastic model of alarm poisoning based on interacting components representing attackers, defenders, alarm infrastructure, and a population of employees. The system is formulated as a continuous-time stochastic process and simulated using the Gillespie Stochastic Simulation Algorithm. Employee behavior is represented through evolving trust and fatigue levels, while technical components evolve through attacker progression, alarm infrastructure compromise, and defender escalation.

Using Monte-Carlo simulation, the study addresses three main questions: how alarm poisoning influences the overall security posture of the system, which patterns of alarm injection most effectively degrade trust and increase fatigue among employees, and how quickly defenders must respond in order to prevent the system from drifting toward widespread distrust and degraded responsiveness.

The remainder of the paper is organized as follows. Section 2 reviews the scientific literature. Section 3 summarizes the theoretical foundations used in the analysis. Section 4 describes the building attack scenario. Section 5 details the stochastic modeling framework and the experimental design. Section 6 examines the simulation results, while Section 7 concludes the paper.

2. Related Work

Research relevant to alarm poisoning spans five main strands: human-factors studies of alarm fatigue, cybersecurity research on alert overload, adversarial work on false data injection and signal manipulation, quantitative models of trust and behavioral adaptation in safety-critical systems, and stochastic Cybersecurity Dynamics approaches for socio-technical and cyber-physical risk analysis.

A substantial human-factors literature has examined how repeated warnings degrade vigilance, trust, and compliance. Le et al. [1] analyze factors shaping employee reactions to fire alarms and show that prior experience and perceived alarm credibility strongly influ-

ence evacuation intentions. In the medical domain, Chromik et al. [2] review computational approaches to mitigating alarm fatigue in intensive care settings, emphasizing filtering, prioritization, and adaptive alarm management. More recently, Michels et al. [3] provide a scoping review of alarm fatigue in healthcare, synthesizing definitions, influencing factors, and mitigation strategies across a broad interdisciplinary literature.

These studies show that excessive warnings can erode attention and weaken confidence in alerting systems. However, they generally interpret alarm fatigue as an unintended consequence of system design or operational overload, rather than as a deliberately exploited attack surface.

A related body of work examines alert overload in cybersecurity monitoring environments. Alahmadi et al. [4] report that large volumes of false positives in security operations centers can reduce analyst confidence in detection systems and promote heuristic filtering of alerts. Other contributions focus on technical mitigation strategies. Yang et al. [5] propose detection architectures aimed at reducing alert volume, while Wang et al. [6] develop reinforcement-learning methods for context-aware alert prioritization in multi-step attack detection. Medvedieva et al. [7] study data augmentation techniques for supervised cyber-attack detection, showing how synthetic data can improve classifier performance under class imbalance.

Although these works seek to improve detection quality and reduce operator workload, they still frame alert overload primarily as an operational burden. In contrast, our work treats sustained false alarm generation as a weaponizable vulnerability that can intentionally degrade organizational responsiveness.

Alarm poisoning is also related to the broader literature on false data injection (FDI) and adversarial manipulation of monitoring signals in control systems, smart grids, and SCADA-like environments. Liu, Ning, and Reiter [8] showed that carefully crafted perturbations of measurement streams can bypass bad-data detection in power-grid state estimation. Xie, Mo, and Sinopoli studied analogous attacks in electricity-market settings and related integrity attacks in power-market operations [9,10]. More recent survey work by Habib et al. [11] synthesizes the challenges, attack surfaces, and research directions associated with false-data-injection attacks in smart-grid cyber-physical systems.

False data injection attacks typically target automated estimation, optimization, or control processes. Alarm poisoning instead targets the human response layer: its purpose is not to corrupt state estimation directly, but to erode trust, increase fatigue, and weaken organizational responsiveness over time. The common structure is deliberate manipulation of monitoring signals in order to alter downstream system behavior; the difference lies in whether the primary target is automated decision logic or human behavior.

Quantitative models of trust evolution have emerged in several safety-critical domains. Song et al. [12] experimentally model trust dynamics in supervised autonomous ship navigation and show how repeated system behavior shapes operator reliance on automation. Although this literature is mainly concerned with benign system error rather than adversarial manipulation, it highlights the importance of modeling trust as a dynamic quantity rather than as a fixed attribute.

Broader work on public warning infrastructures also shows that repeated alerts can influence compliance and risk perception at the population level. Chandra et al. [13] review heat-health warning systems, while Matte et al. [14] discuss the behavioral implications of large-scale warning systems in climate-related contexts. More broadly, research on crisis communication and evacuation decision-making shows that warning response may also be shaped by social influence and peer behavior [15]. In the present work, however, we do not model explicit person-to-person influence. Instead, behavioral change is represented at the population level through alarm exposure, trust, fatigue, and defender posture.

The present work also builds on a methodological strand concerned with stochastic and population-level modeling of cyber–physical and socio–technical systems. Barbierato et al. [16] use Markovian Agents to study crowd safety in enclosed spaces, showing how discrete behavioral states and stochastic transitions can be used to analyze collective safety outcomes under uncertainty. That work is relevant here because it demonstrates how population-level state representations can capture emergent behavioral risk without requiring fully individualized agent-based simulation.

More recent work has extended similar modeling ideas to cybersecurity and infrastructure settings. Cyber threats in autonomous guided vehicles are studied by using mean-field models [17], illustrating how attacker–system interactions can be represented at scale through structured stochastic abstractions. Furthermore, CTMC modeling is combined with geospatial data in [18] to quantify downtime risk in smart-city contexts, showing how stochastic cyber-risk models can be connected to infrastructure-level performance and resilience analysis. At a broader survey level, Xing and Shen [19] review security control approaches for cyber–physical systems under cyber attack, highlighting the importance of linking attack processes, control responses, and resilience-oriented system behavior. These studies are not about alarm fatigue specifically, but they are methodologically relevant because they support the use of stochastic state-based models for cyber–physical degradation processes, resilience metrics, and large-scale socio–technical risk evaluation.

Across these strands, prior studies generally do one of four things: they document alarm fatigue empirically, develop technical mechanisms for reducing alert overload, model adversarial signal manipulation against automated systems, or apply stochastic population-based methods to cyber–physical risk and safety analysis. Much less work integrates attacker behavior, defensive escalation, alarm manipulation, and workforce adaptation within a single stochastic socio–technical framework focused on the degradation of human response quality.

Our contribution differs from prior work in four main ways. First, we treat alarm fatigue as an intentional socio–technical attack vector rather than as an unintended consequence of imperfect alerting systems. Second, unlike false data injection attacks that manipulate automated estimation or control processes, alarm poisoning targets the human response layer by eroding trust and amplifying fatigue. Third, we develop a population-level continuous-time stochastic model that captures nonlinear trust–fatigue dynamics, first-passage collapse behavior, and distributional variability across realizations. Fourth, we combine this behavioral modeling perspective with TRAM agility metrics in order to quantify the temporal competition between attacker escalation and defender mitigation.

3. Scientific Background

This section outlines the theoretical concepts used to formalize the alarm-poisoning scenario introduced in Section 4. We briefly review the Cybersecurity Dynamics perspective, continuous-time stochastic processes, the continuous-time Markov chain formulation adopted in this work, and the resilience metrics used to evaluate attack and defense dynamics.

3.1. Cybersecurity Dynamics Perspective

Cybersecurity Dynamics (CD) views system security as a time-dependent process resulting from the interaction between attackers, defenders, technical infrastructure, and human operators. Instead of treating security as a static property, this perspective models it as an evolving system state influenced by adversarial actions, defensive responses, and behavioral adaptation.

At time t , the global configuration of the system can be represented as

$$S(t) \in \mathcal{S}, \quad (1)$$

where $\mathcal{S}$ denotes a high-dimensional state space describing the operational state of the alarm infrastructure, the progression of the attacker, the posture of the defender organization, the current alarm pressure, and the behavioral state of the workforce.

Within this framework, the overall security condition of the system can be expressed abstractly as

$$\text{situation}(t) = f(V(t), D(t), A(t)), \quad (2)$$

where $V(t)$ represents exploitable vulnerabilities, $A(t)$ denotes attacker capability and activity, and $D(t)$ represents defensive capacity. In practice, the functional form of $f(\cdot)$ is rarely known explicitly, so system behavior is typically explored through stochastic modeling and simulation.

3.2. Continuous-Time Markov Chains

To represent asynchronous interactions between attackers, defenders, infrastructure, and users, the system is modeled as a continuous-time stochastic process. Let $\{X(t)\}_{t \geq 0}$ be a stochastic process defined over a discrete state space $\mathcal{X}$. For small $\Delta t > 0$,

$$\Pr(X(t + \Delta t) = j \mid X(t) = i) = q_{ij}\Delta t + o(\Delta t), \quad (3)$$

where q_{ij} is the transition rate from state i to state j . The matrix $Q = (q_{ij})$ defines a continuous-time Markov chain (CTMC), and the probability distribution $\mathbf{p}(t)$ over the state space evolves according to the Kolmogorov forward equation

$$\frac{d}{dt}p(t) = p(t)Q. \quad (4)$$

CTMC models are particularly suitable for cybersecurity systems because events such as attacks, alarm generation, detection, and mitigation occur at irregular times rather than at fixed time steps.

In the model developed in this work, the global CTMC state includes the attacker stage, alarm infrastructure condition, defender posture, alarm pressure level, and the population distribution of employee behavioral states.

3.3. Population Representation of Workforce Behavior

Human behavior is modeled at the population level rather than at the level of individual agents. Let N denote the number of employees in the organization and let $\mathcal{T}$ denote the set of behavioral states defined by combinations of discrete trust and fatigue levels.

If $n_\ell(t)$ denotes the number of employees in state $\ell \in \mathcal{T}$, the workforce configuration can be summarized by the vector

$$\mathbf{n}(t) = (n_\ell(t))_{\ell \in \mathcal{T}}. \quad (5)$$

Aggregate quantities derived from this distribution provide macroscopic indicators of organizational behavior. For example, the fraction of employees in the lowest trust level can be used as a signal of declining confidence in the alarm infrastructure.

3.4. Agility and Resilience Metrics

System resilience is evaluated using metrics inspired by the TRAM (Trust – Resilience – Agility – Mitigation) framework, which characterizes the temporal competition between at-

tackers and defenders. Let $\tau_k^{(A)}$ and $\tau_m^{(D)}$ denote the times at which attackers and defenders reach successive capability stages. The intervals

$$GT_k^{(A)} = \tau_{k+1}^{(A)} - \tau_k^{(A)}, \quad GT_m^{(D)} = \tau_{m+1}^{(D)} - \tau_m^{(D)} \quad (6)$$

describe the pace at which each side evolves.

Differences in responsiveness can be summarized through measures such as the Lagging-Behind Time

$$LBT = \tau_{\text{target}}^{(D)} - \tau_{\text{target}}^{(A)} \quad (7)$$

which quantifies how long defenders take to reach a mitigation stage after the attacker reaches a corresponding operational stage.

In the alarm-poisoning scenario, alarm injection, defensive mitigation, and workforce adaptation interact through feedback mechanisms. Transition rates depend on system variables such as alarm pressure and defender posture, which can generate nonlinear and stochastic dynamics at the population level. For this reason, probabilistic simulation methods are required to analyze the resilience of the resulting socio-technical system.

4. Case Study: Alarm Poisoning in a Smart Building

We consider a modern office building equipped with an integrated cyber-physical monitoring infrastructure. The system aggregates signals from heterogeneous sensors throughout the facility and distributes alarms both to building occupants and to a centralized security operations center. The building hosts several hundred employees and includes sensitive areas such as server rooms, laboratories, elevators, and access-controlled entrances.

The monitoring platform generates alarms from different operational domains, including fire or smoke alerts, unauthorized access attempts, abnormal server temperatures, suspicious USB activity on workstations, and elevator faults. These alerts are delivered through multiple channels, such as building announcement systems, mobile or workstation notifications, and instructions relayed by floor coordinators. Each alarm category is associated with procedures ranging from routine inspection to immediate evacuation.

Under normal conditions, occupants form expectations about the reliability of the alarm system. Those expectations influence how quickly and seriously alarms are treated. Repeated exposure therefore affects both attention and trust.

In the scenario considered here, an attacker compromises part of the building's cyber infrastructure that interfaces with the alarm dissemination system. The attacker does not immediately disable services or trigger a catastrophic event. Instead, the goal is to weaken confidence in the alarm infrastructure over time by injecting false alerts. We refer to this strategy as alarm poisoning.

Once the monitoring infrastructure is compromised, the attacker can generate alarms of different categories with configurable intensity and relative frequency. Some correspond to safety-critical events such as fire alerts, while others represent lower-severity conditions such as equipment malfunctions or suspicious device activity. By repeatedly injecting misleading alerts, the attacker increases the volume of notifications received by employees and security personnel. Sustained exposure can then degrade attention, encourage skepticism, and reduce compliance with emergency procedures.

In our model, this socio-technical interaction unfolds through four components: an attacker, the alarm infrastructure, the defender organization, and the employee population. The attacker evolves through successive operational stages as the intrusion progresses. The alarm infrastructure can move from a normal state to a compromised one and eventually to a mitigated condition once defensive actions are deployed. The defender organi-

zation reacts to alarm pressure by escalating its operational posture, investigating alerts, and attempting to restore system integrity.

Employees are modeled as a population whose behavioral state evolves over time as alarms recur. The workforce is characterized by two dimensions: a discrete trust level reflecting confidence in the alarm system and a fatigue level representing accumulated exposure to alerts. Alarm events may increase fatigue and affect trust positively when they correspond to genuine incidents or negatively when they are repeatedly revealed to be false. The distribution of trust and fatigue across the workforce therefore changes dynamically as the system operates.

These interactions form a feedback loop between technical infrastructure and human behavior. False alarms increase organizational alarm pressure, accelerate fatigue accumulation, and can trigger defensive escalation. At the same time, repeated false alarms reduce the fraction of employees who fully trust the system, increasing the risk that genuine emergencies will receive slower or weaker responses.

The entities participating in this scenario are summarized in Table 1.

Table 1. Entities participating in the alarm poisoning scenario.

Entity	Role in the Scenario
Attacker	Gradually compromises alarm-related infrastructure and injects false alerts according to configurable strategies.
Employees	Receive alarms and adapt their trust and fatigue levels based on repeated exposure to notifications.
Defender Organization	Monitors alarm activity, escalates defensive posture, investigates incidents, and mitigates compromised components.
Alarm Infrastructure	Aggregates sensor data and distributes alerts; when compromised, it becomes a channel for attacker-generated alarms.

This setting illustrates a socio–technical attack whose primary effects emerge gradually rather than through immediate physical disruption. Manipulation of the alarm channel raises alarm pressure, amplifies fatigue, and progressively erodes trust in the monitoring infrastructure. It therefore provides a concrete setting for studying how alarm poisoning influences collective behavior and how defensive action can mitigate or delay the degradation of organizational responsiveness.

5. Models and Experiments

This section presents the stochastic model used to represent the alarm–poisoning scenario introduced in Section 4 and describes the numerical experiments used to evaluate it. The system is modeled as a continuous–time Markov chain (CTMC) in which attackers, defenders, alarm infrastructure, and employees interact through event-driven transitions.

The global system state is

$$S(t) = (A(t), S_{\text{sys}}(t), D(t), Z(t), n_{T,F}(t)),$$

where

- $A(t)$ denotes the attacker stage,
- $S_{\text{sys}}(t)$ denotes the alarm-infrastructure state,
- $D(t)$ denotes the defender operational posture,
- $Z(t)$ denotes the alarm-pressure variable,
- $n_{T,F}(t)$ denotes the number of employees with trust level T and fatigue level F .

Transitions occur with stochastic rates that depend only on the current configuration of the system. The CTMC is simulated exactly using the Gillespie Stochastic Simulation Algorithm (SSA), which samples event times and event types without discretizing time.

5.1. System Components

5.1.1. Attacker

The attacker evolves through four capability levels,

$$A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3,$$

representing external presence, foothold, operational control, and full alarm-injection capability. The corresponding transitions are

$$A_0 \xrightarrow{\lambda_{01}} A_1, \quad A_1 \xrightarrow{\lambda_{12}} A_2, \quad A_2 \xrightarrow{\lambda_{23}} A_3.$$

5.1.2. Alarm Infrastructure

The alarm infrastructure evolves through three states,

$$S_0 : \text{Normal}, \quad S_1 : \text{Compromised}, \quad S_2 : \text{Mitigated}.$$

Once the attacker has obtained at least a foothold, the alarm system can become compromised:

$$S_0 \xrightarrow{\kappa_{\text{comp}}} S_1.$$

When defensive mitigation is active, the system can transition to a mitigated state:

$$S_1 \xrightarrow{\kappa_{\text{mit}}(D)} S_2.$$

5.1.3. Defender Organization

The defender evolves through three operational postures,

$$D_0 : \text{Monitoring}, \quad D_1 : \text{Investigating}, \quad D_2 : \text{Mitigating}.$$

Escalation depends on alarm pressure:

$$D_0 \xrightarrow{\gamma_{01}(Z)} D_1, \quad D_1 \xrightarrow{\gamma_{12}(Z)} D_2.$$

Relaxation occurs only when alarm pressure has dropped sufficiently:

$$D_2 \xrightarrow{\gamma_{20}} D_0.$$

5.1.4. Employees

Employees are represented at the population level through occupancy counts over trust and fatigue compartments:

$$T \in \{0, 1, 2\}, \quad F \in \{0, \dots, F_{\max}\},$$

where T denotes trust in the alarm system and F denotes accumulated fatigue. The model tracks

$$n_{T,F}(t),$$

the number of employees occupying each trust–fatigue compartment at time t .

5.2. Alarm Processes

Two classes of alarm events are modeled.

True alarms.

True alarms correspond to genuine incidents such as fires, intrusions, or equipment failures. For each alarm type j , true alarms occur with baseline hazard rate λ_j .

Fake alarms.

Fake alarms are generated by the attacker once the alarm infrastructure is compromised and the attacker has reached operational control. Fake alarms occur with overall intensity Λ_{fake} and are distributed across alarm categories according to a strategy-specific mixture π_j .

Each alarm event increments the alarm-pressure variable:

$$Z \rightarrow \min(Z + 1, Z_{\max}).$$

Alarm pressure then decays through a linear death process:

$$Z \rightarrow Z - 1 \quad \text{at rate } \omega Z.$$

5.3. Trust and Fatigue Dynamics

When an alarm of type j occurs, each employee is exposed with probability ρ_j . Exposure increases fatigue according to

$$(T, F) \rightarrow (T, \min(F + 1, F_{\max})).$$

Fatigue also decreases spontaneously through recovery events,

$$(T, F) \rightarrow (T, F - 1),$$

with per-individual rate

$$\phi F.$$

Trust dynamics depend on whether the alarm is fake or genuine. For fake alarms, trust can decrease:

$$(T, F) \rightarrow (T - 1, F), \quad T > 0.$$

For genuine alarms, trust can increase:

$$(T, F) \rightarrow (T + 1, F), \quad T < 2.$$

More specifically, after a fake alarm of type j , the probability that an employee in compartment (T, F) loses one trust level is

$$p_{\downarrow}(j, T, F, D) = \min\left\{1, \alpha_{\downarrow}(j) g_f(F) h_D^{\downarrow}(D) q_T^{\downarrow}(T)\right\},$$

where

$$g_f(F) = 1 + g_f^{\text{slope}} F.$$

After a genuine alarm of type j , the probability that an employee in compartment (T, F) gains one trust level is

$$p_{\uparrow}(j, T, F, D) = \min\left\{1, \alpha_{\uparrow}(j) \tilde{g}_f(F) h_D^{\uparrow}(D) q_T^{\uparrow}(T)\right\},$$

where

$$\tilde{g}_f(F) = \frac{1}{1 + \tilde{g}_f^{\text{slope}} F}.$$

In words, trust erosion and trust recovery are both shaped by four factors:

1. the alarm type j through $\alpha_{\downarrow}(j)$ and $\alpha_{\uparrow}(j)$,
2. the current fatigue level F through $g_f(F)$ and $\tilde{g}_f(F)$,
3. the defender posture D through $h_D^{\downarrow}(D)$ and $h_D^{\uparrow}(D)$,
4. the current trust level T through $q_T^{\downarrow}(T)$ and $q_T^{\uparrow}(T)$.

The defender-posture multipliers are intended to represent the assumption that visible organizational response can partially buffer trust erosion after false alarms and can more strongly support trust restoration after genuine alarms. Accordingly, higher defender postures reduce the probability of trust loss and increase the probability of trust recovery. These values should be interpreted as scenario-based behavioral coefficients rather than empirically calibrated constants.

In other words, prompt and visible defensive action is assumed to reassure employees more effectively than inaction, thereby making genuine alarms more trust-restoring and false alarms less trust-damaging.

The employee-state dynamics can be summarized through four elementary transition rules, reported in Table 2. Alarm exposure increases fatigue with probability ρ_j when an alarm of type j occurs, while fatigue recovery is modeled as a spontaneous per-individual transition with rate ϕF . Trust degradation is triggered by fake alarms and trust restoration by genuine alarms; in both cases, the corresponding transition probabilities depend explicitly on alarm type, fatigue level, defender posture, and current trust level through the functions $p_{\downarrow}(j, T, F, D)$ and $p_{\uparrow}(j, T, F, D)$.

It can be seen that fatigue amplifies trust degradation after fake alarms, while fatigue attenuates trust restoration after genuine alarms. The numerical values used in the simulations are listed in Table 3.

Table 2. Summary of employee-state transition rules.

Event Type	State Transition	Governing Quantity	Conditions
Alarm exposure	$(T, F) \rightarrow (T, \min(F + 1, F_{\max}))$	ρ_j	alarm of type j occurs
Fatigue recovery	$(T, F) \rightarrow (T, F - 1)$	ϕF	$F > 0$
Trust degradation	$(T, F) \rightarrow (T - 1, F)$	$p_{\downarrow}(j, T, F, D)$	fake alarm, $T > 0$
Trust restoration	$(T, F) \rightarrow (T + 1, F)$	$p_{\uparrow}(j, T, F, D)$	genuine alarm, $T < 2$

Table 3. Parameter values used in the CTMC simulation model. Rates are expressed per hour. The fake-alarm intensity is held fixed across strategies, so differences between attacker strategies arise from the alarm-type mixture rather than total fake-alarm volume.

Symbol/Item	Value	Meaning
$N_{\text{employees}}$	200	Number of employees
$F_{\max}$	6	Maximum fatigue level
$Z_{\max}$	12	Maximum alarm-pressure level
Z_{relax}	2	Defender relaxation threshold
Δ	0.05 h	Response-time deadline (3 min)
$quorum_m$	8	Quorum responders for evacuation metric
Evacuation alarm types	fire, intruder	Alarm classes used for evacuation-related interpretation

Table 3. Cont.

Symbol/Item	Value	Meaning
Attacker progression parameters		
λ_{01}	0.030 h^{-1}	Attacker progression $A_0 \rightarrow A_1$
λ_{12}	0.020 h^{-1}	Attacker progression $A_1 \rightarrow A_2$
λ_{23}	0.050 h^{-1}	Attacker progression $A_2 \rightarrow A_3$
κ_{comp}	0.020 h^{-1}	Alarm-system compromise rate $S_0 \rightarrow S_1$
Defender and alarm-pressure parameters		
ω	0.10 h^{-1}	Alarm-pressure decay coefficient in rate ωZ
$\gamma_{01}^{\text{base}}$	0.010 h^{-1}	Baseline defender escalation $D_0 \rightarrow D_1$
$\gamma_{01}^{\text{slope}}$	0.015 h^{-1}	Alarm-pressure sensitivity for $D_0 \rightarrow D_1$
$\gamma_{12}^{\text{base}}$	0.005 h^{-1}	Baseline defender escalation $D_1 \rightarrow D_2$
$\gamma_{12}^{\text{slope}}$	0.010 h^{-1}	Alarm-pressure sensitivity for $D_1 \rightarrow D_2$
γ_{20}	0.020 h^{-1}	Defender relaxation rate $D_2 \rightarrow D_0$
$\kappa_{\text{mit}}(D_0)$	0.005 h^{-1}	Mitigation rate in posture D_0
$\kappa_{\text{mit}}(D_1)$	0.012 h^{-1}	Mitigation rate in posture D_1
$\kappa_{\text{mit}}(D_2)$	0.020 h^{-1}	Mitigation rate in posture D_2
Fake-alarm process		
Λ_{fake}	1.0 h^{-1}	Overall fake-alarm intensity (fixed across strategies)
π_{fire}	(0.60, 0.15, 0.10, 0.10, 0.05)	Fire–Spam mixture
π^{low}	(0.01, 0.04, 0.15, 0.35, 0.45)	Low–Severity Spam mixture
π^{mixed}	(0.05, 0.15, 0.35, 0.25, 0.20)	Mixed-strategy mixture
Fatigue and trust parameters		
ϕ	0.06 h^{-1}	Fatigue recovery coefficient
g_f^{slope}	0.25	Fatigue amplification for trust degradation
$\tilde{g}_f^{\text{slope}}$	0.15	Fatigue attenuation for trust recovery
$h_D^{\downarrow}(D_0)$	1.00	Defender multiplier for trust decrease
$h_D^{\downarrow}(D_1)$	0.75	Defender multiplier for trust decrease
$h_D^{\downarrow}(D_2)$	0.55	Defender multiplier for trust decrease
$h_D^{\uparrow}(D_0)$	1.00	Defender multiplier for trust increase
$h_D^{\uparrow}(D_1)$	1.25	Defender multiplier for trust increase
$h_D^{\uparrow}(D_2)$	1.45	Defender multiplier for trust increase
$q_T^{\downarrow}(2)$	1.00	Trust-level multiplier for trust decrease from high trust
$q_T^{\downarrow}(1)$	1.10	Trust-level multiplier for trust decrease from medium trust
$q_T^{\downarrow}(0)$	0.00	No lower trust level below $T = 0$
$q_T^{\uparrow}(0)$	1.00	Trust-level multiplier for trust increase from low trust
$q_T^{\uparrow}(1)$	0.85	Trust-level multiplier for trust increase from medium trust
$q_T^{\uparrow}(2)$	0.00	No higher trust level above $T = 2$

5.4. CTMC State Transition Structure

Figure 1 illustrates the principal transition structure of the CTMC governing attacker progression, alarm-system compromise, defender escalation, and alarm-pressure dynamics. Figure 2 depicts the employee trust–fatigue dynamics.

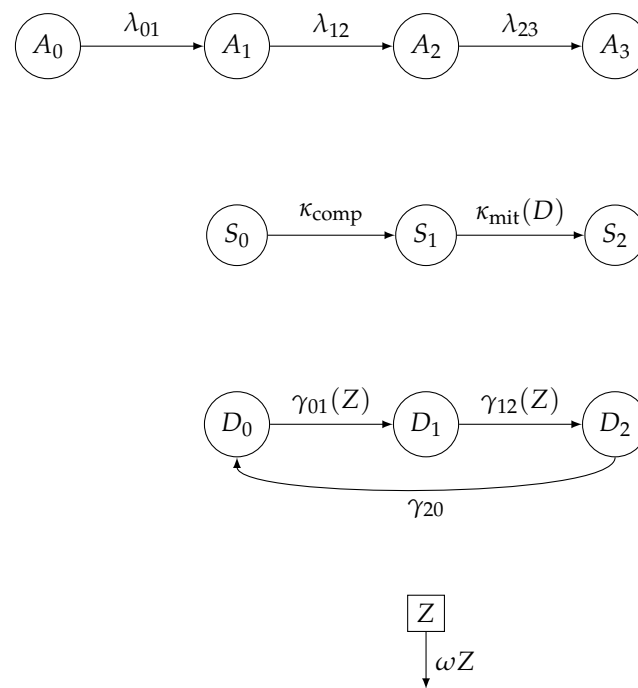


Figure 1. Main CTMC state transitions for attacker progression, alarm-infrastructure compromise, defender escalation, and alarm-pressure dynamics.

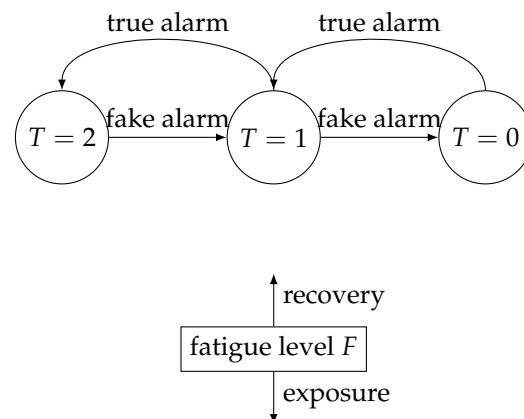


Figure 2. Conceptual transitions governing employee trust and fatigue dynamics. Fake alarms decrease trust, genuine alarms can restore trust, alarm exposure increases fatigue, and recovery events decrease fatigue.

5.5. Stochastic Simulation

The system is simulated using the Gillespie Stochastic Simulation Algorithm [20]. At each step, the algorithm:

1. computes the rates of all enabled events given the current state,
2. samples the waiting time to the next event from an exponential distribution,
3. selects the next event proportionally to its rate,
4. applies the corresponding state update.

This procedure generates exact sample paths of the underlying CTMC. Because the model evolves through discrete alarm, mitigation, trust-shift, fatigue-recovery, and escalation events in a finite population, and because the quantities of interest are first-passage times and distributional outcomes rather than only mean behavior, an event-driven SSA formulation is more appropriate than a deterministic mean-field approximation.

Although the abstract state space is combinatorially large, the simulation does not enumerate it explicitly. Instead, each realization is generated by evaluating only the transition intensities active at the current state. Table 3 lists the parameter values used in the numerical experiments.

5.6. Experimental Campaign

We considered three attacker strategies. In all cases, the overall fake-alarm intensity was held fixed at

$$\Lambda_{\text{fake}} = 1.0 \text{ h}^{-1},$$

so that strategies differ only in their alarm-type composition and not in total fake-alarm volume.

The three strategy mixtures are:

$$\pi^{\text{fire}} = (0.60, 0.15, 0.10, 0.10, 0.05),$$

$$\pi^{\text{low}} = (0.01, 0.04, 0.15, 0.35, 0.45),$$

$$\pi^{\text{mixed}} = (0.05, 0.15, 0.35, 0.25, 0.20),$$

in the alarm-type order

(fire, intruder, server_temp, usb_malware, elevator).

Thus:

- Fire–Spam emphasizes safety-critical fire alarms;
- Low–Severity Spam concentrates on nuisance and lower-severity alarms;
- Mixed uses an intermediate composition, weighted toward operational and medium-severity alarm categories rather than a purely balanced allocation.

For each strategy, we executed 50 independent Monte Carlo simulations with different random seeds over a time horizon of 168 h, for a total of 150 CTMC realizations.

The analysis focused on four classes of macroscopic observables:

- time to collapse, defined in the baseline analysis as the first time at which the low-trust fraction exceeds 0.40;
- fatigue accumulation over time;
- alarm-pressure trajectories $Z(t)$;
- defender agility metrics, especially the TRAM Lagging-Behind Time (LBT).

The structural form of the model is informed by the alarm-fatigue, warning-response, and cybersecurity-monitoring literature, whereas the numerical coefficients in Table 3 should be interpreted primarily as scenario-based exploratory values unless otherwise stated.

All simulations and analysis scripts used to generate the figures in this paper are available in the public repository https://github.com/EBarbierato/ciberattack_fatigue (accessed on 27 April 2026).

5.7. Computational Burden and Replication Sufficiency

To assess computational tractability, we recorded per-run wall-clock times and SSA event counts. Table 4 summarizes the computational burden of the simulation campaign in terms of wall-clock time and SSA event counts across strategies.

Table 4. Computational summary of the Monte Carlo campaign. For each attacker strategy, the table reports the number of runs, wall-clock execution time per realization, and the number of SSA events processed per realization. These values provide an empirical characterization of computational tractability under the chosen parameterization.

Strategy	Runs	Median Wall Time (s)	Min (s)	Max (s)	Median SSA Events	Min/Max SSA Events
Fire–Spam	50	0.0291	0.0037	0.0973	2864.5	426/9771
Low–Severity Spam	50	0.0266	0.0038	0.0752	2532.0	426/7075
Mixed	50	0.0271	0.0039	0.0916	2656.5	426/8848

To justify the use of 50 replications per strategy, we examined how the median and interquartile range (IQR) of the main outcome statistics stabilized as the number of runs increased. Figure 3 reports these replication-sufficiency diagnostics for time-to-collapse and LBT.

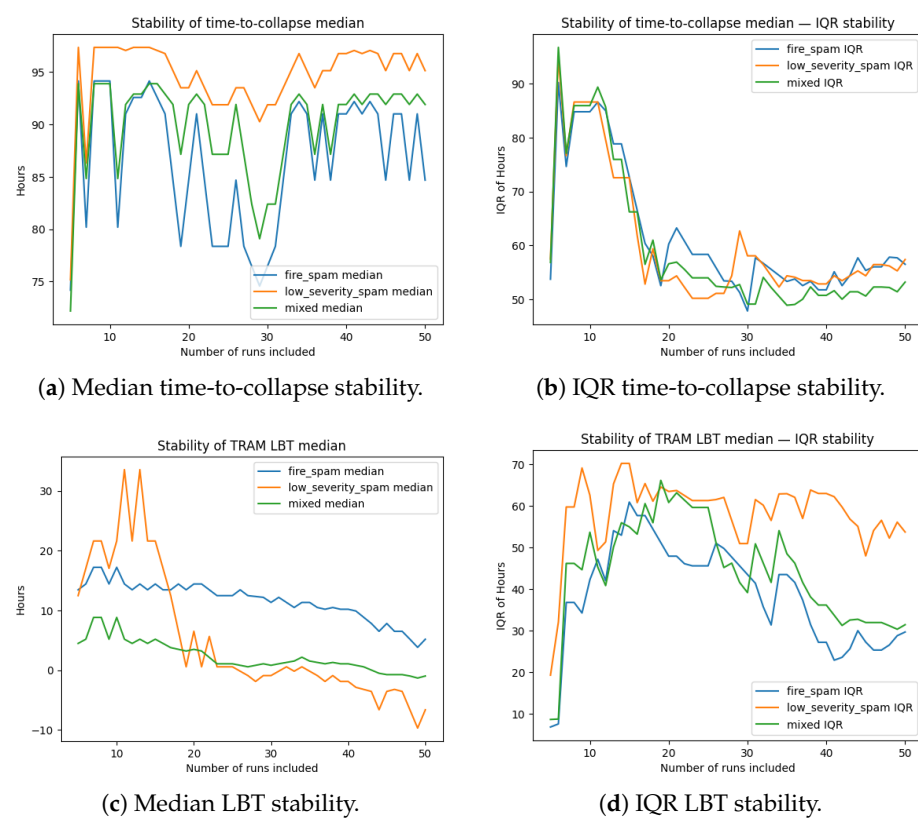


Figure 3. Replication-sufficiency diagnostics for the Monte Carlo campaign. The panels show how the median and interquartile range (IQR) of time-to-collapse and TRAM LBT evolve as the number of replications increases. Median-based comparisons become substantially more stable after roughly 25–35 runs, while dispersion estimates also become markedly less volatile by 50 runs. On this basis, 50 replications per strategy were retained as a reasonable compromise between computational cost and distributional stability.

5.8. Sensitivity, Threshold Robustness, and Ablation Analysis

To assess the robustness of the results to the scenario-based parameterization, we performed a one-at-a-time sensitivity analysis around the baseline values reported in Table 3. We perturbed eight key parameters by factors of 0.75, 1.00, and 1.25 while holding all other parameters fixed: the overall fake-alarm intensity Λ_{fake} , the fatigue recovery rate ϕ , the fatigue amplification and attenuation coefficients in the trust dynamics, the defender escalation slopes, the mitigation rate in posture D_2 , and the fire-specific trust erosion

coefficient. For each perturbed setting and each attacker strategy, we executed 30 Monte Carlo replications over the same 168-h horizon used in the main experiments.

The sensitivity analysis focused on the same core outcome measures as the main study: median and interquartile range of time-to-collapse, the fraction of runs crossing the collapse threshold within the simulation horizon, and median and interquartile range of the TRAM Lagging-Behind Time (LBT). Overall, the main qualitative ranking of strategies was robust across most tested settings: in 22 of the 24 parameter-setting combinations, Fire–Spam remained the strategy with the shortest median time-to-collapse, Mixed remained intermediate, and Low–Severity Spam remained the slowest. The strongest sensitivity was observed for the fatigue recovery rate ϕ , which produced the largest shifts in median time-to-collapse, whereas the fatigue-attenuation coefficient for trust recovery had negligible effect over the tested range. Table 5 summarizes the main robustness patterns.

Table 5. Summary of the one-at-a-time sensitivity analysis. Across the 24 perturbed parameter-setting combinations (eight parameters, two non-baseline perturbations, evaluated against the baseline ordering), the default median collapse-time ordering was preserved in 22 cases. The strongest sensitivity was observed for the fatigue recovery rate ϕ , while the fatigue attenuation coefficient for trust recovery had negligible influence over the tested range.

Parameter	Main Effect on Outcomes	Ranking Preserved?	Notes
ϕ (fatigue recovery rate)	Strongest sensitivity for median time-to-collapse; maximum shift of about 38% across tested settings	No	At $\phi = 0.75 \times$ baseline, the default collapse-time ordering is altered
$\alpha_{\downarrow}$ (fire)	Moderate-to-strong sensitivity of collapse time; directly affects Fire–Spam effectiveness	No	At $1.25 \times$ baseline, the default collapse-time ordering is altered
$\gamma_{01}^{\text{slope}}$	Moderate sensitivity of both collapse time and LBT	Yes	Defender escalation from monitoring to investigation affects both behavioral degradation and lag
$\gamma_{12}^{\text{slope}}$	Moderate sensitivity, especially for LBT	Yes	Faster escalation from investigation to mitigation improves defender timing
Λ_{fake}	Moderate sensitivity of collapse time and LBT	Yes	Higher fake-alarm intensity shifts outcomes in the expected direction without changing the main strategy ranking
g_f^{slope}	Mild-to-moderate sensitivity of collapse time	Yes	Fatigue amplification of trust loss matters, but less than ϕ
$\kappa_{\text{mit}}(D_2)$	Mild sensitivity, primarily through defender timing	Yes	Mitigation speed at the highest defender posture has limited but visible influence
$\bar{g}_f^{\text{slope}}$	Negligible effect over the tested range	Yes	Fatigue attenuation of trust recovery is nearly inert under $\pm 25\%$ perturbations

We also examined the robustness of the collapse criterion itself by repeating the time-to-collapse analysis for low-trust thresholds of 0.30, 0.40, and 0.50. The qualitative ordering of strategies remained unchanged across all three thresholds: Fire–Spam always produced the shortest median time-to-collapse, Low–Severity Spam the longest, and Mixed remained intermediate. The same ordering was preserved for the number of non-collapse runs within the 168-h horizon, with Fire–Spam consistently showing the fewest non-collapse runs and Low–Severity Spam the most. These results indicate that the main conclusions are not artifacts of the particular baseline choice of 0.40, although the absolute collapse times and non-collapse frequencies vary with the threshold.

Finally, to isolate the roles of the main feedback mechanisms, we conducted an ablation analysis in which selected couplings were removed from the model while all other components were retained. Three ablations were examined: (i) removal of alarm-pressure feedback in defender escalation, (ii) removal of defender escalation itself, and (iii) removal of the trust-fatigue coupling in the trust-transition probabilities. Figure 4 summarizes the resulting changes in median time-to-collapse, non-collapse runs, and collapse rate across attacker strategies. Relative to the baseline model, removing alarm-pressure feedback reduced the separation among attacker strategies by shifting the median time-to-collapse upward and compressing the differences between Fire-Spam, Mixed, and Low-Severity Spam. Removing defender escalation increased collapse frequency across all strategies, indicating that escalation acts as a protective mechanism even when it does not fully prevent collapse. Removing trust-fatigue coupling weakened the effectiveness of alarm poisoning, especially for Low-Severity Spam and Mixed, showing that fatigue-mediated trust dynamics are an important amplification mechanism in the full model. These ablation results support the interpretation that alarm pressure, defender escalation, and trust-fatigue coupling each play distinct causal roles in shaping the collapse behavior observed in the baseline simulations.

A complementary global robustness analysis based on Latin hypercube sampling was conducted over eight jointly varied parameters, each sampled over a multiplicative range of 0.75–1.25 around the baseline. Across 100 sampled parameter sets, the strict baseline ordering of median time-to-collapse (Fire-Spam fastest, Mixed intermediate, Low-Severity Spam slowest) was preserved in 21% of cases. Fire-Spam remained the fastest-collapse strategy in 37% of sampled parameter sets, while Low-Severity Spam remained the slowest in 43%. Thus, under broader joint uncertainty, the comparative ranking of attacker strategies is not globally stable. This result reinforces the interpretation of the model as a mechanism-exploration framework rather than a deployment-specific predictive ranking model, while still supporting the broader conclusion that alarm composition, defender escalation, and trust-fatigue coupling materially shape collapse behavior.

As a supplementary heterogeneity-oriented extension, we repeated the simulation campaign with two employee classes characterized by differentiated alarm exposure profiles: a regular class comprising 80% of the workforce and a high-exposure class comprising the remaining 20%, with the latter assigned a $1.5\times$ exposure multiplier relative to the baseline per-alarm exposure probabilities. The resulting behavior is summarized in Figure 5. At the aggregate level, the qualitative ranking remains similar to the baseline scenario: Fire-Spam still produces the shortest median time-to-collapse, Low-Severity Spam the longest, and Mixed remains intermediate. However, the class-specific outcomes are not identical. Under Fire-Spam, the regular class ends with a slightly higher median low-trust fraction than the high-exposure class, whereas under Low-Severity Spam the high-exposure class shows the larger final low-trust fraction. Under Mixed, both classes converge to nearly the same high level of degradation. These results suggest that limited workforce heterogeneity can shift where trust degradation concentrates, even when the aggregate ordering of strategies remains qualitatively similar. In this sense, differentiated exposure profiles do not overturn the main mechanism-level conclusions of the model, but they do show that subgroup-specific stress concentrations can emerge even under a relatively simple two-class extension.

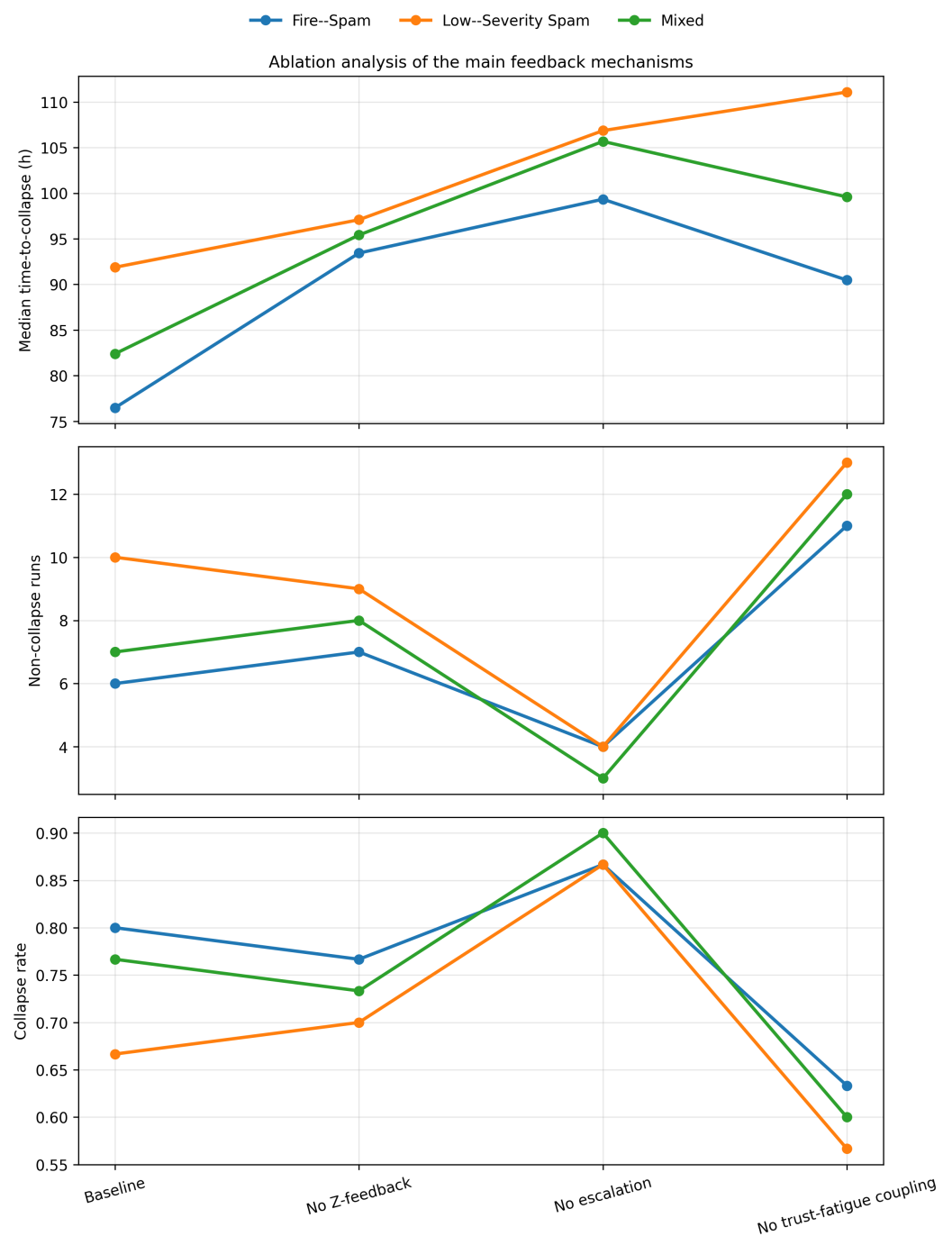


Figure 4. Ablation analysis of the main feedback mechanisms. The top panel shows median time-to-collapse, the middle panel depicts the number of non-collapse runs, and the bottom panel indicates the collapse rate across ablation conditions. Removing alarm-pressure feedback compresses the separation among attacker strategies, removing defender escalation increases collapse frequency, and removing trust-fatigue coupling weakens the behavioral impact of alarm poisoning.

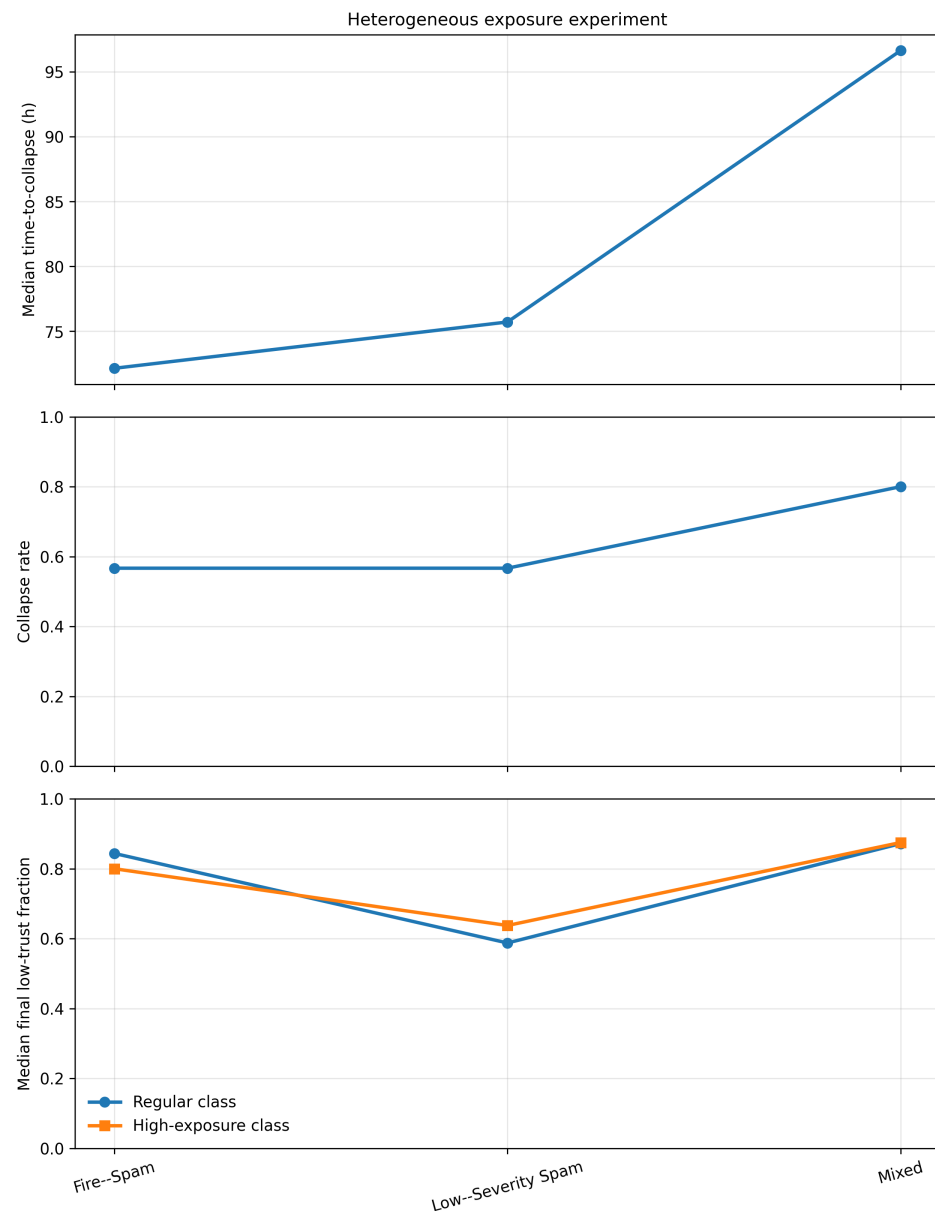


Figure 5. Supplementary heterogeneous-exposure experiment with two employee classes: a regular class and a high-exposure class. The top panel shows median time-to-collapse, the middle panel depicts the collapse rate, and the bottom panel shows the median final low-trust fraction by employee class. The extension shows that differentiated exposure can produce subgroup-specific differences in trust degradation while preserving the broader mechanism-level role of alarm composition and defensive timing.

6. Discussion

This section discusses the Monte–Carlo results introduced in Section 5. The analysis moves from time-dependent variables to collapse timing and defender agility. Because all three attacker strategies use the same overall fake-alarm intensity, $\Lambda_{\text{fake}} = 1.0 \text{ h}^{-1}$, the differences below isolate alarm-type composition rather than total false-alarm volume.

6.1. Trust Dynamics Across Attacker Strategies

Figure 6a shows a clear ordering in the low-trust fraction. *Fire–Spam* drives the fastest increase and pushes the system toward widespread distrust soon after the attacker reaches its most disruptive stage. *Mixed* shows a similar but delayed transition, whereas *Low–Severity Spam* requires longer exposure.

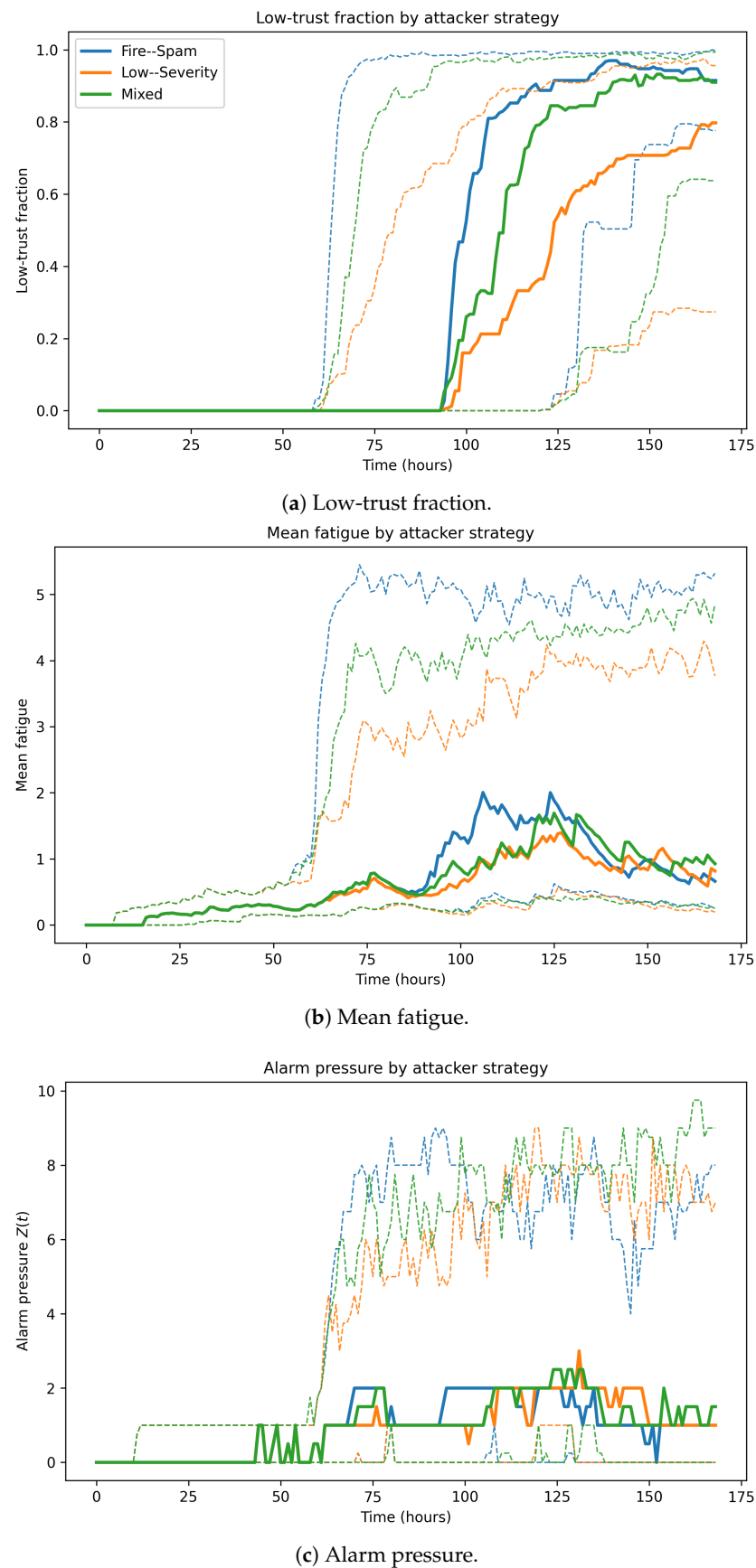


Figure 6. Ensemble time-dependent dynamics under the three attacker strategies. Solid lines denote the median across Monte Carlo runs; dashed lines denote the first and third quartiles. Panel (a) shows the low-trust fraction, panel (b) mean fatigue, and panel (c) alarm pressure.

This indicates that repeated false alarms with high perceived severity are stronger behavioral disruptors than nuisance-dominated campaigns, even when total fake-alarm intensity is fixed. In the model, the effect arises from the interaction among alarm type, trust erosion, fatigue accumulation, and defender posture rather than from alarm count alone.

6.2. Fatigue Accumulation

Figure 6b shows the evolution of mean fatigue. High-severity alarm injection again has the strongest effect, producing sustained fatigue over much of the campaign. *Mixed* also yields elevated fatigue, though with a slower rise. By contrast, *Low-Severity Spam* produces slower accumulation and allows partial recovery when alarm pressure declines.

Together with the trust trajectories, these results show that workforce behavior depends not only on the number of alarms but also on their perceived urgency. Repeated exploitation of safety-critical alarm categories therefore alters response conditions more strongly than equally frequent nuisance-oriented alarm streams.

6.3. Alarm Pressure as a System Variable

Figure 6c reports the alarm-pressure variable $Z(t)$, which captures the accumulated intensity of recent alarm activity. Across strategies, periods of rapid distrust growth coincide with increases in alarm pressure. This suggests pressure regimes in which the system becomes more fragile: once $Z(t)$ is sufficiently high, trust degrades faster and fatigue remains elevated longer. *Fire-Spam* reaches these regimes more often than nuisance-dominated campaigns.

Alarm pressure is therefore a useful early-warning variable. Although it is not itself a behavioral state, it can precede or accompany the degradation of trust and responsiveness. From a defensive perspective, sustained excursions of $Z(t)$ above nominal levels—for example relative to the defender relaxation threshold Z_{relax} —could trigger automated review, temporary alert filtering, or earlier escalation. Such cues would not replace human judgment, but they could reduce time spent in a lagging defensive posture.

6.4. Stochastic Time-to-Collapse

Figure 7 provides distributional evidence on collapse timing, where “collapse” denotes an operational first-passage threshold rather than irreversible organizational failure. Among runs reaching the threshold within 168 h, median time-to-collapse was 84.68 h for *Fire-Spam* (IQR 59.18–115.68 h), 95.13 h for *Low-Severity Spam* (IQR 66.69–124.08 h), and 91.90 h for *Mixed* (IQR 63.64–116.84 h). The threshold was not reached in 8/50, 14/50, and 11/50 runs, respectively.

These counts are important substantively. In particular, *Low-Severity Spam* fails to induce threshold crossing in a relatively large fraction of runs, indicating that nuisance-dominated alarm poisoning is less reliable as a behavioral degradation mechanism even when it still accumulates fatigue.

Because a non-negligible fraction of runs did not collapse within the simulation horizon, we also used Kaplan–Meier estimates with right-censoring at 168 h (Figure 8). Estimated median survival times were 96.99 h for *Fire-Spam*, 118.13 h for *Low-Severity Spam*, and 107.59 h for *Mixed*, preserving the same descriptive ordering. However, an omnibus log-rank test did not detect statistically significant separation among the survival curves ($\chi^2 = 3.03$, $p = 0.220$), and pairwise comparisons with Holm correction were likewise non-significant.

The 0.40 threshold should be read as a scenario-based marker of widespread distrust, not a universal empirical failure boundary. Repeating the analysis for thresholds of 0.30, 0.40, and 0.50 leaves the qualitative ordering unchanged: *Fire-Spam* collapses fastest, *Low-Severity Spam* slowest, and *Mixed* remains intermediate. Figure 9 therefore shows

that the main conclusions are not artifacts of the baseline threshold, even though absolute collapse times and non-collapse frequencies vary.

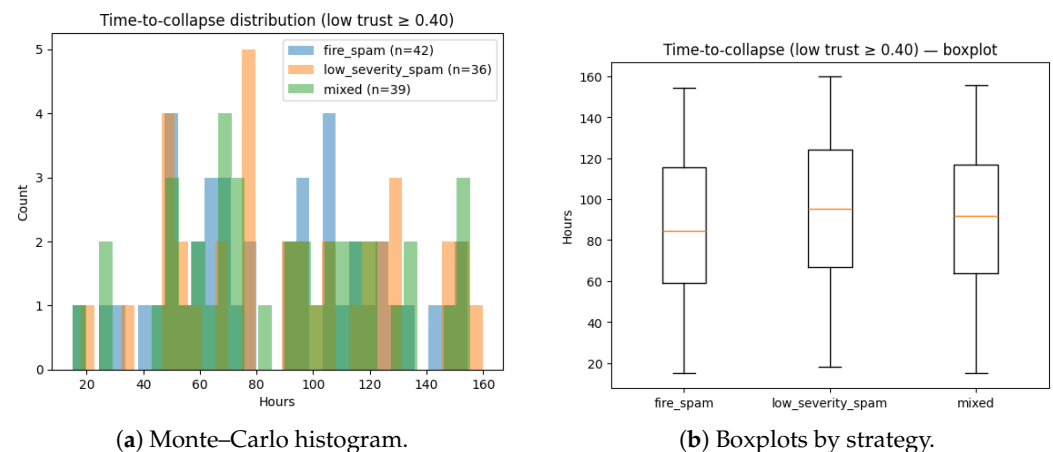


Figure 7. Distribution of time to collapse, defined as the first time at which the low-trust fraction exceeds 0.40.

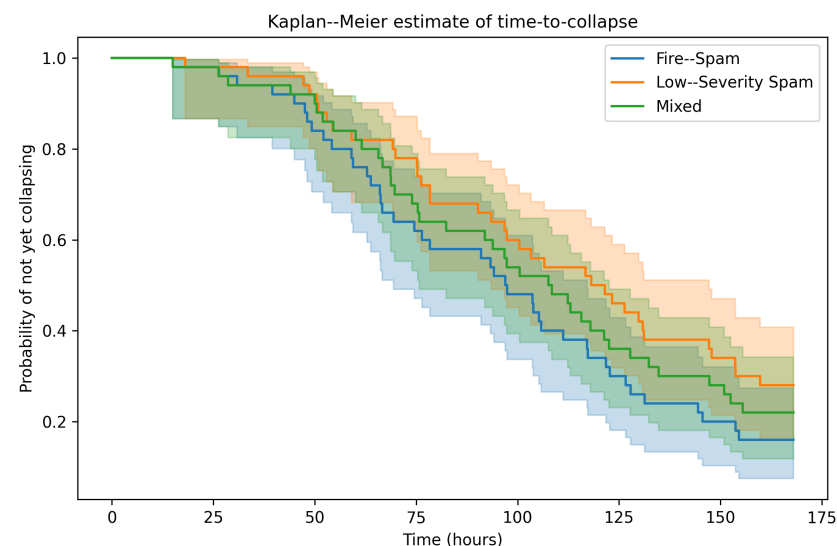


Figure 8. Kaplan–Meier estimates of time-to-collapse, where collapse is defined as the first time at which the low-trust fraction exceeds 0.40. Runs that do not reach the threshold within 168 h are treated as right-censored observations.

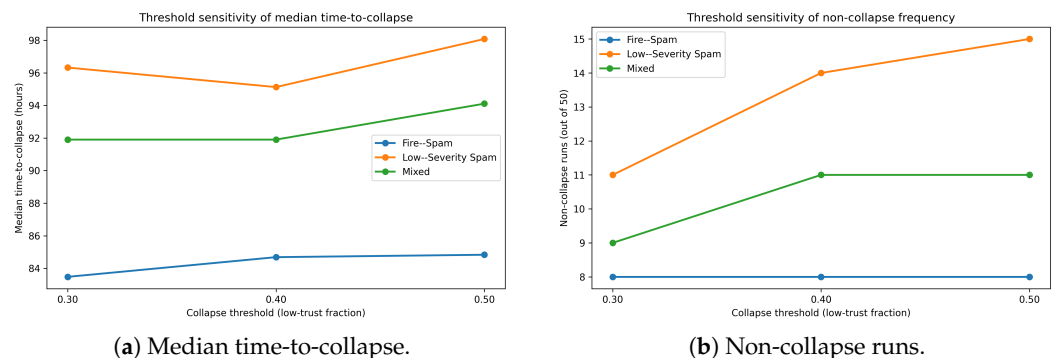


Figure 9. Sensitivity of the collapse analysis to the low-trust threshold. The qualitative ordering of attacker strategies is preserved across thresholds of 0.30, 0.40, and 0.50: Fire–Spam always produces the fastest collapse and the fewest non-collapse runs, while Low–Severity Spam remains the slowest and least reliable.

Taken together, the main evidence for strategic differences lies in descriptive ordering, non-collapse frequencies, and robustness patterns rather than strong inferential separation at the present sample size. More broadly, the results confirm that alarm poisoning produces inherently stochastic outcomes: attacker strategy shifts the distribution of collapse times, but identical parameter settings can still yield materially different realized trajectories. Alarm-poisoning risk should therefore be interpreted probabilistically rather than deterministically.

6.5. Defender Agility and Lag Distributions

Figure 10 reports the empirical distributions of the TRAM Lagging-Behind Time (LBT). The histogram shows overall spread and asymmetry, while the boxplots summarize medians, interquartile ranges, and extremes by strategy.

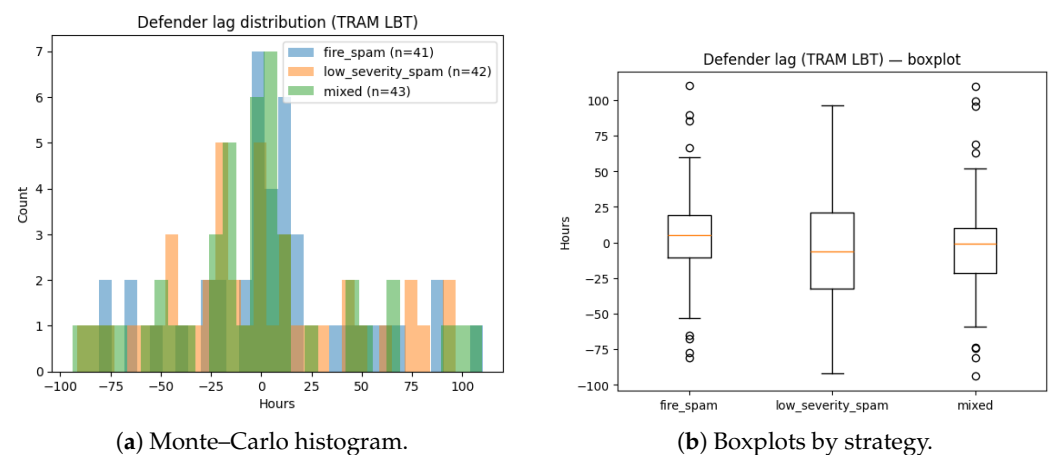


Figure 10. Distribution of the TRAM Lagging-Behind Time (LBT), defined as the time difference between the defender reaching mitigation posture and the attacker reaching its highest operational stage. Positive values indicate that the defender lags behind the attacker; negative values indicate that the defender reaches mitigation earlier.

The three attacker strategies produce different lag profiles. *Fire-Spam* tends to yield positive LBT values, meaning that mitigation often arrives only after the attacker has entered its most disruptive stage. *Low-Severity Spam* more often yields negative LBT values, meaning that defender escalation can occur before full attacker impact. *Mixed* remains closer to zero but still shows wide variability.

Operationally, a positive LBT means mitigation is late relative to attacker escalation: the building remains exposed, false alarms continue to accumulate, and trust and fatigue can deteriorate further before the defender regains control. A negative LBT means mitigation arrives before the attacker reaches its most damaging phase, increasing the chance that distrust remains contained.

What matters, therefore, is not simply whether the defender reacts, but whether escalation and mitigation occur on time scales comparable to attacker progression. In alarm-poisoning scenarios, a few hours can separate a localized disturbance from broader organizational degradation. These results underscore the importance of defensive agility: delay is costly because it sustains alarm pressure and worsens the trust-fatigue state of the workforce.

6.6. Synthesis and Cybersecurity Dynamics Implications

The results reveal a common mechanism linking technical and behavioral dynamics. Repeated false alarms increase alarm pressure, which accelerates fatigue accumulation and gradually shifts the employee population toward lower trust levels. Once the low-trust

fraction becomes sufficiently large, the system enters a regime of degraded responsiveness in which genuine alarms are less behaviorally effective.

The one-at-a-time sensitivity analysis confirms that the main qualitative findings are robust across most tested perturbations. In 22 of the 24 perturbed parameter-setting combinations, the median time-to-collapse ordering remained unchanged, with Fire–Spam producing the fastest collapse, Mixed remaining intermediate, and Low–Severity Spam remaining the slowest. The strongest sensitivity was observed for the fatigue recovery rate ϕ , whereas the fatigue attenuation coefficient governing trust recovery had negligible influence over the tested range.

From a Cybersecurity Dynamics perspective, several properties stand out. First, the model is strongly nonlinear: alarm pressure, fatigue, trust, and defender posture interact to produce threshold-like changes rather than smooth proportional responses. Second, the system is intrinsically stochastic: even under fixed parameters, realizations can diverge substantially in both collapse timing and defender lag. Third, the workforce acts as a large population-level subsystem whose aggregate state can change abruptly once enough employees occupy low-trust and high-fatigue compartments.

The results support the view that alarm poisoning is a credible socio-technical attack vector. They also show that defender agility—measured on the scale of hours rather than days—plays a central role in maintaining trust and preserving effective emergency response under sustained alarm manipulation.

The present model assumes a behaviorally homogeneous workforce at the population-compartment level and does not represent explicit role heterogeneity or person-to-person influence. Extending the framework to heterogeneous employee classes and social interaction networks therefore remains an important direction for future work.

From an operational perspective, these findings suggest several design implications. Alarm infrastructures should be treated not only as safety mechanisms but also as potential attack surfaces, so alarm integrity, segmentation, and access control deserve the same attention as other mission-critical cyber–physical assets. Alarm-pressure indicators such as $Z(t)$ may serve as early-warning signals for escalation workflows, automated review, or temporary filtering, especially when false alarms accumulate before more visible behavioral degradation appears. Defender lag metrics such as LBT can likewise help organizations assess whether their response procedures are fast enough relative to plausible attacker progression. At the same time, these implications should be interpreted as qualitative design insights rather than universal policy thresholds. The present study supports rapid escalation, alarm monitoring, and adaptive response, but it does not establish a single empirically calibrated operational rule for all buildings or organizations.

To clarify the scope of the claims made in this paper, it is useful to distinguish three levels of interpretation. First, the simulation results establish several validated findings within the model: under the stated assumptions and parameterizations, alarm composition, defender escalation, trust–fatigue coupling, and exposure heterogeneity all materially affect collapse behavior and timing. Second, the model suggests plausible operational hypotheses, such as the usefulness of alarm pressure as an early-warning signal, the importance of limiting defender lag, and the possibility that highly exposed subgroups may act as early indicators of degradation. Third, the paper points to forward-looking design recommendations, including adaptive alarm management, organization-specific escalation thresholds, richer heterogeneous workforce models, and empirically calibrated policy design.

7. Conclusions

This paper examined alarm poisoning as a socio–technical cyber–physical attack on smart-building alarm infrastructures. The study asked how repeated false alarms alter

employee behavior, which attacker strategies are most disruptive, and whether defensive agility can limit the resulting loss of responsiveness.

To address these questions, we developed a stochastic Cybersecurity Dynamics model in which attackers, defenders, alarm infrastructure, and employees interact through a continuous-time Markov chain simulated with the Gillespie Stochastic Simulation Algorithm. Employee behavior was represented through population-level trust and fatigue compartments, allowing repeated alarm exposure to generate feedback between technical signaling and human response.

The results answer the three research questions directly. First, repeated false alarms can degrade response conditions by increasing fatigue, reducing trust, and pushing the system toward threshold-like declines in responsiveness. Second, under the baseline scenario, strategies concentrated on high-severity alarms tend to produce faster and more reliable degradation than nuisance-dominated campaigns: Fire–Spam generally yields the shortest collapse times, Mixed remains intermediate, and Low–Severity Spam is slower and less reliable. Third, defensive agility matters substantially: faster escalation and mitigation reduce defender lag, limit alarm-pressure accumulation, and improve the chances of containing degradation before low trust becomes widespread.

These findings must nevertheless be interpreted with caution. Replication-sufficiency diagnostics supported the baseline Monte–Carlo campaign, and local robustness checks showed that the qualitative ordering of strategies is preserved across most one-at-a-time perturbations and across alternative collapse thresholds. Ablation analysis further showed that alarm-pressure feedback, defender escalation, and trust–fatigue coupling each contribute distinctly to collapse behavior, while a supplementary heterogeneous-exposure extension showed that limited workforce heterogeneity can shift where degradation concentrates.

At the same time, the model does not support a strong claim of universal comparative dominance. The survival analysis did not show strong inferential separation among strategies at the present sample size, and the Latin hypercube robustness analysis showed that the baseline ranking is not globally stable under joint parameter variation. The comparative results should therefore be interpreted as probabilistic tendencies within a scenario-based mechanism-exploration framework rather than as fixed predictive rankings for all real deployments.

From an operational perspective, the study highlights alarm infrastructures as meaningful attack surfaces in cyber–physical environments. The model suggests that alarm pressure, trust distribution, and defender lag are potentially useful monitoring variables, and that sustained increases in alarm pressure may provide an early signal for escalation or automated review. These implications are best understood as plausible operational hypotheses and design priorities, not as universally calibrated policy thresholds.

The present analysis remains intentionally simplified. It considers a single building configuration, relies on scenario-based rather than empirically calibrated coefficients, and represents behavior primarily at the population-compartment level. Future work should therefore focus on empirical grounding through alarm logs, expert elicitation, or behavioral data; richer heterogeneous workforce models; broader global uncertainty analysis; and adaptive defenses that combine human response, automation, and alarm-management policy. Within those limits, the results support the view that alarm poisoning is a credible socio–technical attack vector whose consequences depend on the interaction between technical compromise and human adaptation.

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